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UNIVERSITY OF GHANA

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DEPARTMENT OF TEACHER EDUCATION

SCHOOL OF EDUCATION AND LEADERSHIP

COLLEGES OF EDUCATION

END OF SEMESTER ONE EXAMINATIONS, 2024/2025

TIME: 2 HOURS

PEKI COLLEGE OF EDUCATION LIBRARY PKCE

COURSE CODE: TEEG 311

COURSE TITLE: Introduction to Early Grade Science III

INSTRUCTIONS: This paper consists of **Sections A and B**. Answer all the questions in Section A and **THREE** from Section B.

SECTION A
[25 marks]
Answer all the questions in this section.

1. What is the primary function of a plant's roots?
   A) Photosynthesis
   B) Water absorption
   C) Reproduction
   D) Seed dispersal

2. Which of the following is a habitat for aquatic plants?
   A) Desert
   B) Ocean
   C) Grassland
   D) Tundra

3. Fill in the blank: The process by which plants reproduce is called _____.
   A) Germination
   B) Photosynthesis
   C) Propagation
   D) Transpiration

4. What is the first stage of germination?
   A) Leaf development
   B) Root emergence
   C) Seed swelling
   D) Flowering

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5. Which of the following classes do mammals belong to?
   A) Reptilia
   B) Amphibia
   C) Aves
   D) Mammalia

6. Fill in the blank: The three states of matter are solid, liquid, and _____.
   A) Plasma
   B) Gas
   C) Dust
   D) Vapour

7. What element is essential for plant photosynthesis?
   A) Nitrogen
   B) Carbon
   C) Oxygen
   D) Hydrogen

8. Which soil type holds the most water?
   A) Sandy soil
   B) Clay soil
   C) Loamy soil
   D) Silt soil

9. Fill in the blank: The primary gas that plants take in during photosynthesis is _____.
   A) Carbon dioxide
   B) Oxygen
   C) Nitrogen
   D) Helium

10. Which of the following tools is used to measure temperature?
    A) Barometer
    B) Thermometer
    C) Hydrometer
    D) Celsius

11. What is the typical volume measurement for liquids?
    A) Kilograms
    B) Meters
    C) Liters
    D) Centimetres

12. Which of the following is NOT a characteristic of air?
    A) It has mass
    B) It is visible
    C) It occupies space
    D) It can exert pressure

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13. Fill in the blank: The process by which animals are classified is called _____.
    A) Taxonomy
    B) Ecology
    C) Biology
    D) Anatomy

14. What is the role of student teachers in promoting diversity?
    A) To enforce uniformity
    B) To share knowledge and perspectives
    C) To eliminate differences
    D) To create barriers

15. Which plant part is primarily responsible for photosynthesis?
    A) Roots
    B) Stem
    C) Leaves
    D) Flowers

16. What is the main purpose of soil?
    A) To hold water only
    B) To support plant growth
    C) To provide shelter for animals
    D) To store minerals only

17. Fill in the blank: The temperature at which water freezes is _____ degrees Celsius.
    A) 0
    B) 32
    C) 100
    D) -1

18. What is the term for the movement of water through a plant?
    A) Absorption
    B) Transpiration
    C) Evaporation
    D) Propagation

19. Which type of measurement is used for volume?
    A) Kilograms
    B) Meters
    C) Liters
    D) Centimetres

20. Fill in the blank: The classification system for animals is called _____.
    A) Biology
    B) Botany
    C) Zoology
    D) Taxonomy

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21. What is the primary gas produced by plants during photosynthesis?
    A) Oxygen
    B) Carbon dioxide
    C) Nitrogen
    D) Methane

22. Which of the following is a key factor affecting plant growth?
    A) Light
    B) Temperature
    C) Water
    D) All of the above

23. Which of the following is NOT a method of measuring temperature?
    A) Thermocouple
    B) Thermometer
    C) Barometer
    D) Infrared sensor

24. Which of the following is a characteristic of air?
    A) It is a solid
    B) It has no mass
    C) It can be compressed
    D) It is always warm

25. Fill in the blank: The primary role of a student teacher in a diverse classroom is to _____.
    A) Enforce rules
    B) Promote inclusivity
    C) Limit discussions
    D) Standardize learning

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SECTION B

[75 marks]

Answer THREE questions from this section

Q1 (a) i Discuss the importance of plant habitats in maintaining biodiversity. **4 marks**

ii Explain the process of germination **4 marks**

iii What is the significance of germination in plant life cycles? **2 marks**

(b) i What are the main characteristics used to classify animals? **6 marks**

ii Describe the water cycle in an annotated diagram. **6 marks**

iii What is the importance of the water cycle to plant and animal life? **3 marks**

Q2 (a) Explain the role of student teachers in fostering environmental awareness among young learners. **6 marks**

(b) i Explain **THREE** factors that contribute to the classification of animal **6 marks**

ii Why is the classification of animals important? **3 marks**

(c) i Discuss **THREE** soil conservation practices. **6 marks**

ii Evaluate how soil conservation practices can enhance sustainability in agriculture. **4 marks**

Q3 (a) i What is the role of soil in supporting ecosystems? **3 marks**

ii In an annotated diagram, explain the main components of soil. **6 marks**

iii How do the components of soil support plant growth? **4 Marks**

(b) i Describe the process of photosynthesis. **4 marks**

ii What is the importance of photosynthesis to life on Earth? **2 marks**

iii Discuss the significance of temperature and moisture in the germination process. **6 Marks**

Q4 (a) i Explain **THREE** methods of plant propagation and state **ONE** **8 Marks**

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advantage for each method of propagation.

ii What is air quality? How does it affect plant health and growth. **4 Marks**

iii Differentiate between the **TWO** types of germination. **2 Marks**

(b) i Explain **TWO** strategies that student teachers can use to promote diversity in the classroom. **4 Marks**

ii What are the common instruments used to measure Temperature, Mass, time, and Volume? **4 Marks**

iii Differentiate between Mass and Weight. **4 Marks**

Q5 (a) i What is the relationship between animals and their habitats? **4 marks**

ii Describe **FOUR** human activities that impact air quality and consequently plant health **8 marks**

iii Explain the role of ecosystems in the water cycle. **4 marks**

(b) i Discuss the concept of biodiversity and its importance in maintaining ecological balance. **4 Marks**

ii Discuss the impact of climate change on plant habitats and biodiversity. **5 Marks**